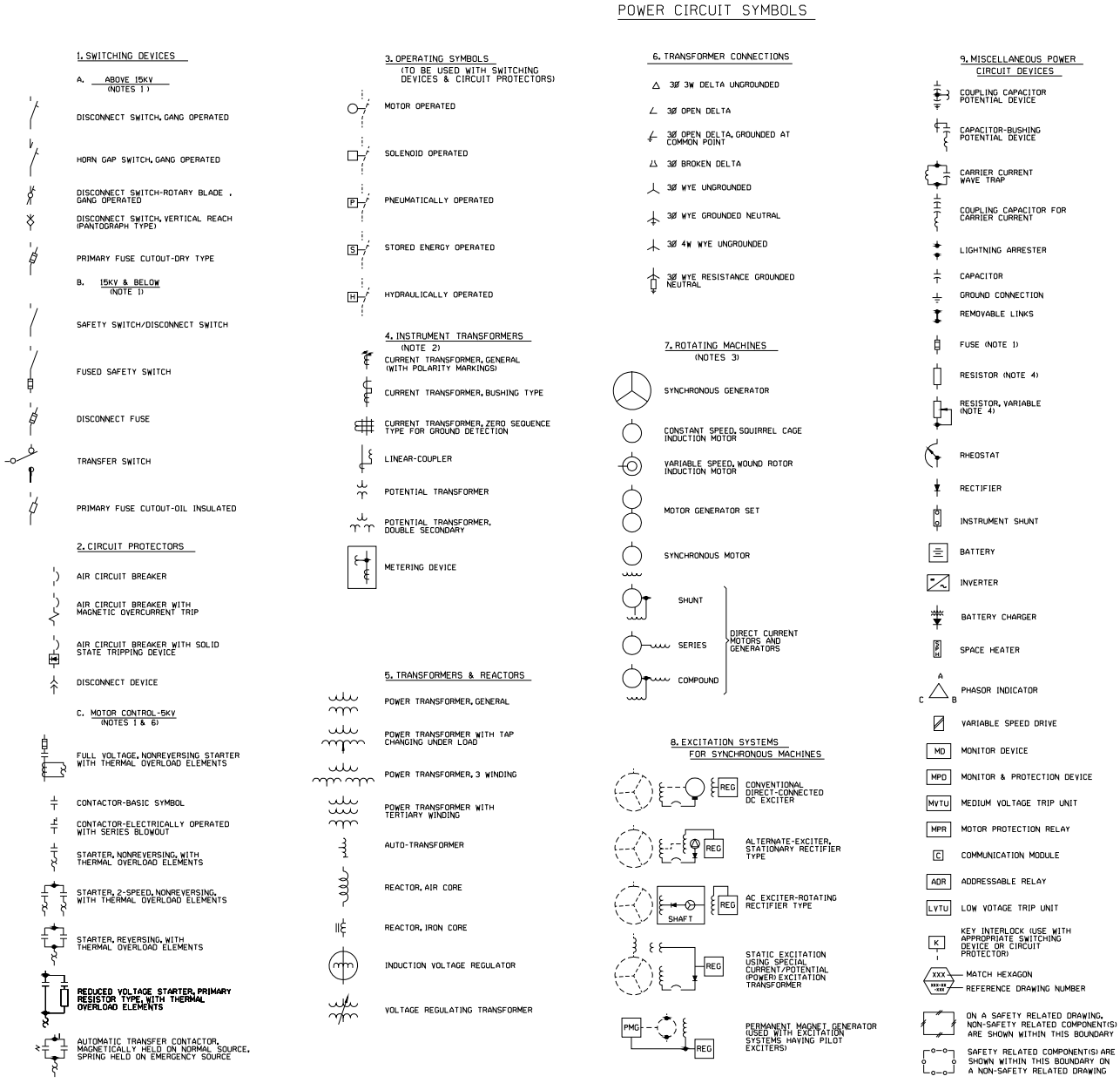
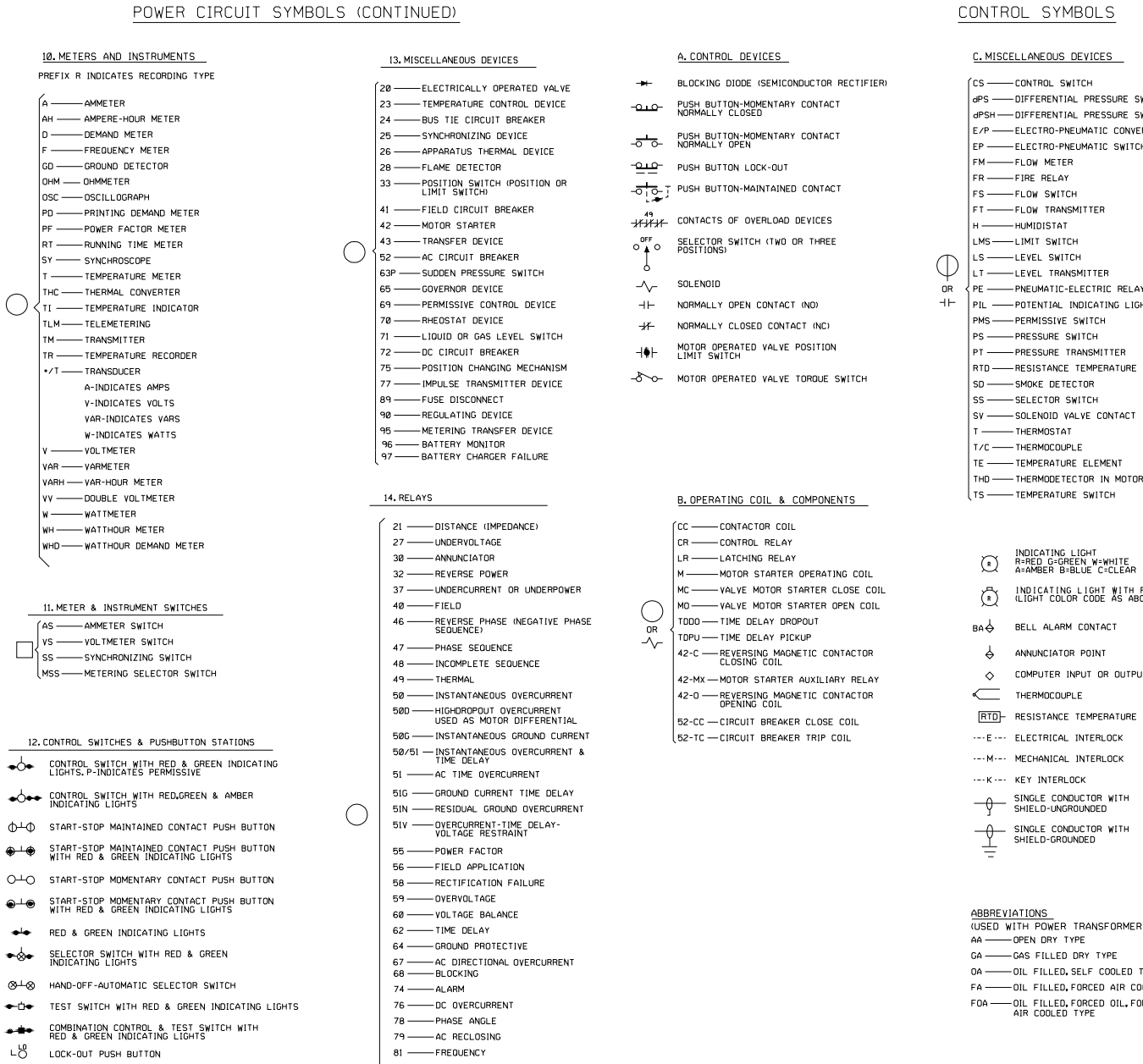
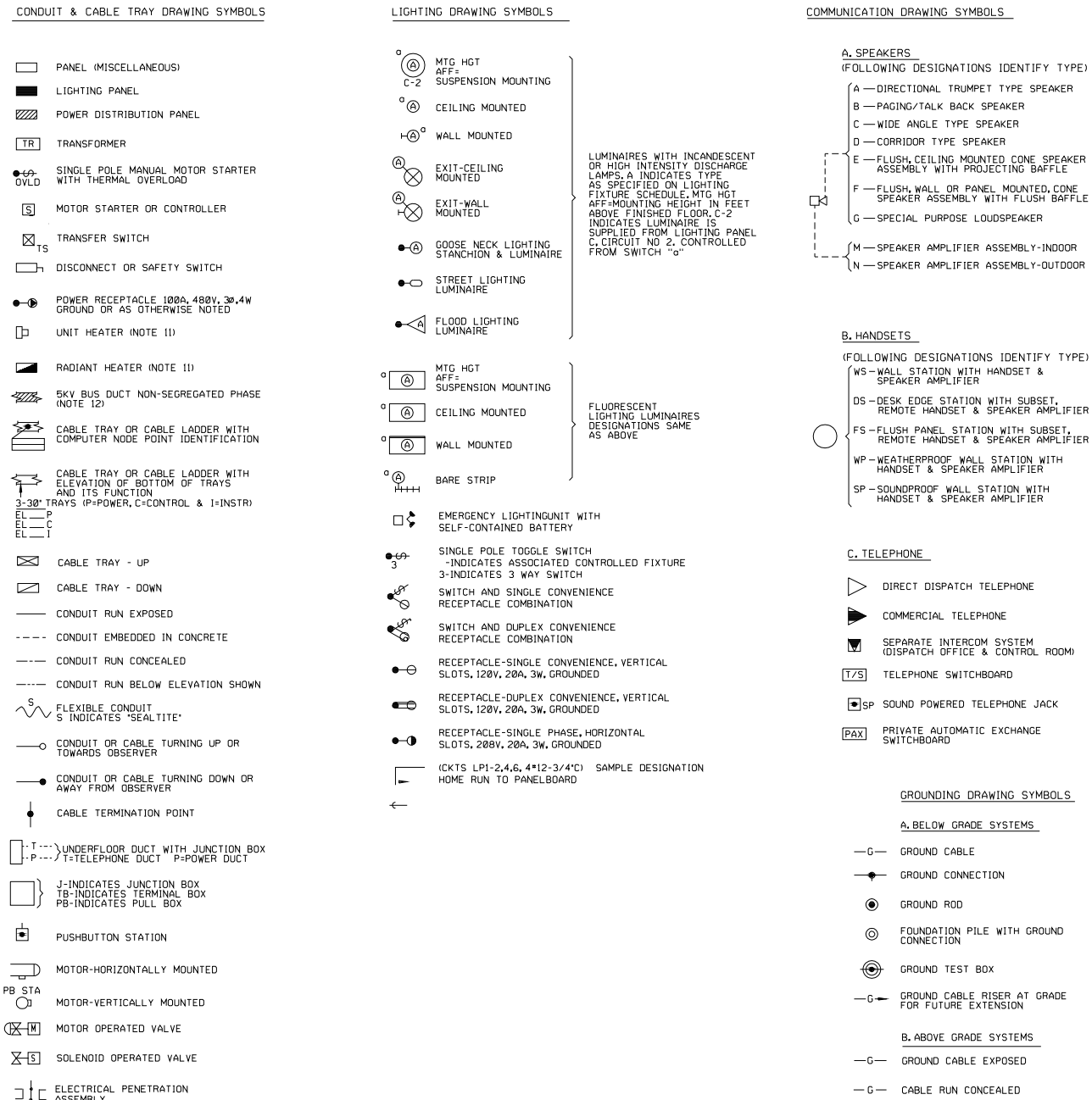
# 1. Introduction and General Description of the Plant

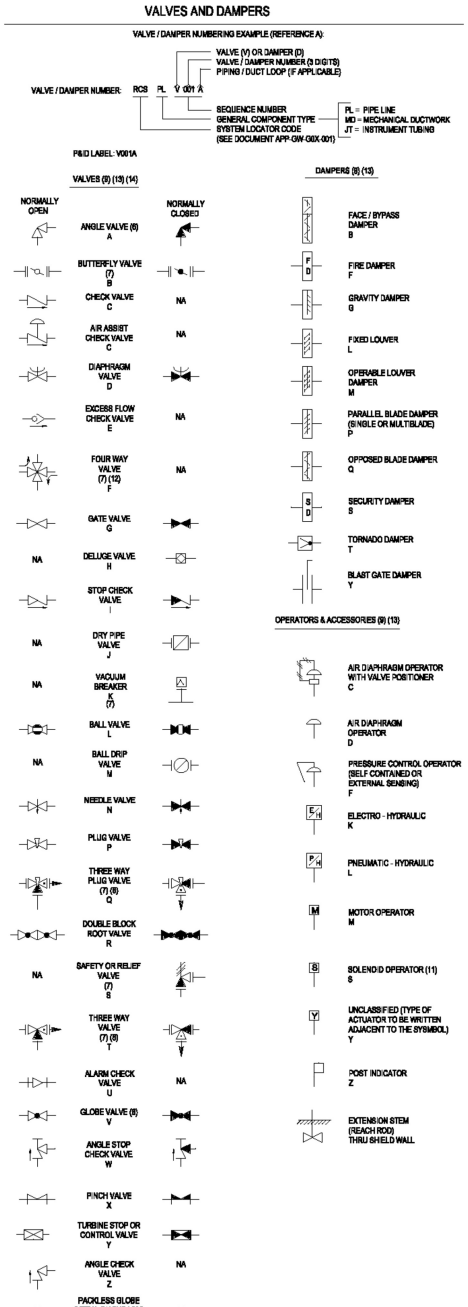


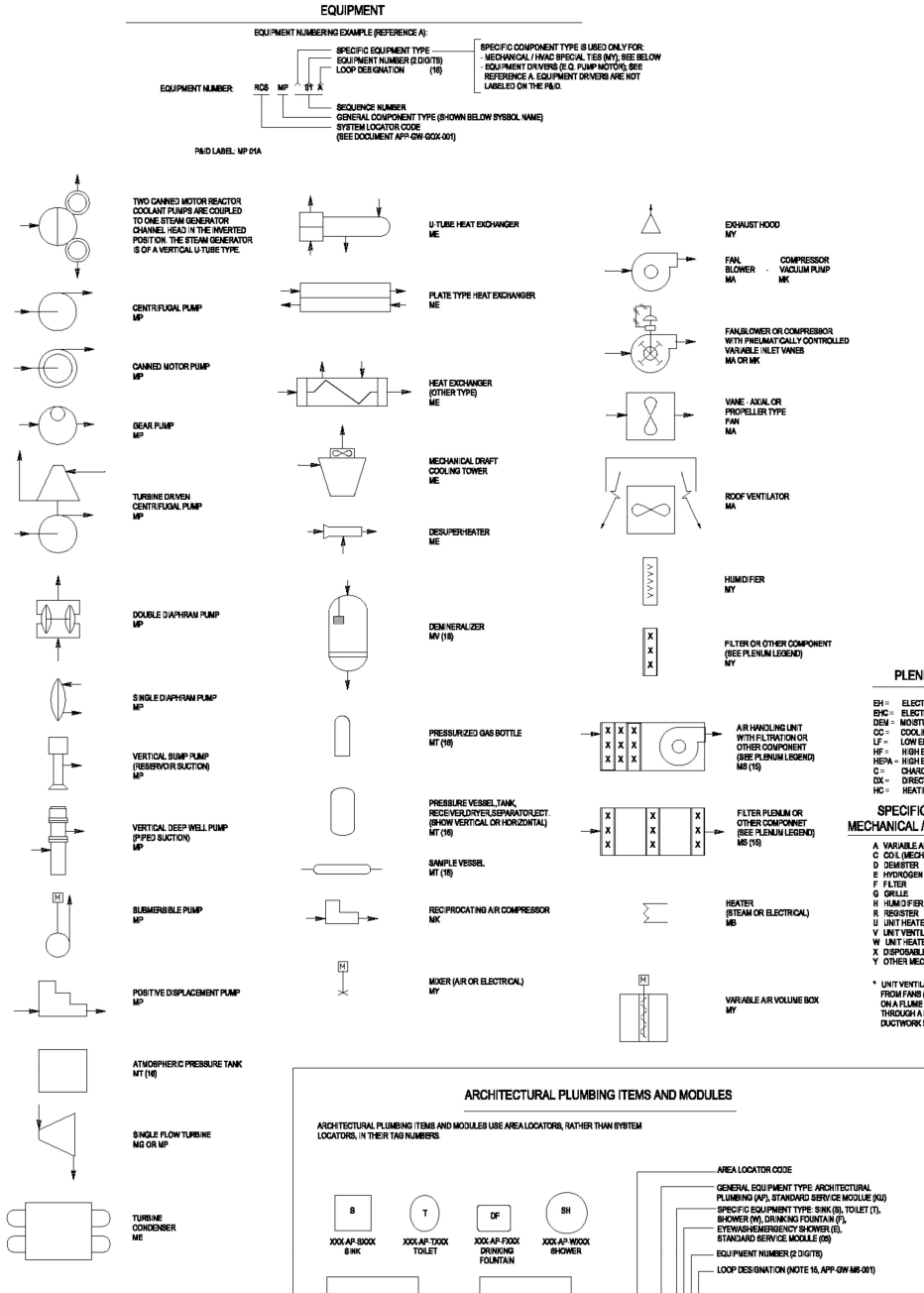

POWER CIRCUIT SYMBOLS (CONTINUED)
10. METERS AND INSTRUMENTS
PREFIX R INDICATES RECORDING TYPE
A — AMMETER
AH — AMPERE-HOUR METER
D — DEMAND METER
F — FREQUENCY METER
GD — GROUND DETECTOR
OHM — OHMMETER
OSC — OSCILLOGRAPH
PD — PRINTING DEMAND METER
PF — POWER FACTOR METER
RT — RUNNING TIME METER
SY — SYNCHROSCOPE
T — TEMPERATURE METER
THC — THERMAL CONVERTER
TI — TEMPERATURE INDICATOR
TLM — TELEMETERING
TM — TRANSMITTER
TR — TEMPERATURE RECORDER
•/T — TRANSDUCER
A-INDICATES AMPS
V-INDICATES VOLTS
VAR-INDICATES VARS
W-INDICATES WATTS
V — VOLTMETER
VAR — VARMETER
VARH — VAR-HOUR METER
VV — DOUBLE VOLTMETER
W — WATTMETER
WH — WATTHOUR METER
WHD — WATTHOUR DEMAND METER
11. METER & INSTRUMENT SWITCHES
AS — AMMETER SWITCH
VS — VOLTMETER SWITCH
SS — SYNCHRONIZING SWITCH
MSS — METERING SELECTOR SWITCH
12. CONTROL SWITCHES & PUSHBUTTON STATIONS
CONTROL SWITCH WITH RED & GREEN INDICATING LIGHTS. P-INDICATES PERMISSIVE
CONTROL SWITCH WITH RED,GREEN & AMBER INDICATING LIGHTS
START-STOP MAINTAINED CONTACT PUSH BUTTON
START-STOP MAINTAINED CONTACT PUSH BUTTON WITH RED & GREEN INDICATING LIGHTS
START-STOP MOMENTARY CONTACT PUSH BUTTON
START-STOP MOMENTARY CONTACT PUSH BUTTON WITH RED & GREEN INDICATING LIGHTS
RED & GREEN INDICATING LIGHTS
SELECTOR SWITCH WITH RED & GREEN INDICATING LIGHTS
HAND-OFF-AUTOMATIC SELECTOR SWITCH
TEST SWITCH WITH RED & GREEN INDICATING LIGHTS
COMBINATION CONTROL & TEST SWITCH WITH RED & GREEN INDICATING LIGHTS
LOCK-OUT PUSH BUTTON
13. MISCELLANEOUS DEVICES
20 — ELECTRICALLY OPERATED VALVE
23 — TEMPERATURE CONTROL DEVICE
24 — BUS TIE CIRCUIT BREAKER
25 — SYNCHRONIZING DEVICE
26 — APPARATUS THERMAL DEVICE
28 — FLAME DETECTOR
33 — POSITION SWITCH (POSITION OR LIMIT SWITCH)
41 — FIELD CIRCUIT BREAKER
42 — MOTOR STARTER
43 — TRANSFER DEVICE
52 — AC CIRCUIT BREAKER
63P — SUDDEN PRESSURE SWITCH
65 — GOVERNOR DEVICE
69 — PERMISSIVE CONTROL DEVICE
70 — RHEOSTAT DEVICE
71 — LIQUID OR GAS LEVEL SWITCH
72 — DC CIRCUIT BREAKER
75 — POSITION CHANGING MECHANISM
77 — IMPULSE TRANSMITTER DEVICE
89 — FUSE DISCONNECT
90 — REGULATING DEVICE
95 — METERING TRANSFER DEVICE
96 — BATTERY MONITOR
97 — BATTERY CHARGER FAILURE
14. RELAYS
21 — DISTANCE (IMPEDANCE)
27 — UNDERVOLTAGE
30 — ANNUNCIATOR
32 — REVERSE POWER
37 — UNDERCURRENT OR UNDERPOWER
40 — FIELD
46 — REVERSE PHASE (NEGATIVE PHASE SEQUENCE)
47 — PHASE SEQUENCE
48 — INCOMPLETE SEQUENCE
49 — THERMAL
50 — INSTANTANEOUS OVERCURRENT
50D — HIGHDROPOUT OVERCURRENT USED AS MOTOR DIFFERENTIAL
50G — INSTANTANEOUS GROUND CURRENT
50/51 — INSTANTANEOUS OVERCURRENT & TIME DELAY
51 — AC TIME OVERCURRENT
51G — GROUND CURRENT TIME DELAY
51N — RESIDUAL GROUND OVERCURRENT
51V — OVERCURRENT-TIME DELAY-VOLTAGE RESTRAINT
55 — POWER FACTOR
56 — FIELD APPLICATION
58 — RECTIFICATION FAILURE
59 — OVERVOLTAGE
60 — VOLTAGE BALANCE
62 — TIME DELAY
64 — GROUND PROTECTIVE
67 — AC DIRECTIONAL OVERCURRENT
68 — BLOCKING
74 — ALARM
76 — DC OVERCURRENT
78 — PHASE ANGLE
79 — AC RECLOSING
81 — FREQUENCY
CONTROL SYMBOLS
A. CONTROL DEVICES
BLOCKING DIODE (SEMICONDUCTOR RECTIFIER)
PUSH BUTTON-MOMENTARY CONTACT NORMALLY CLOSED
PUSH BUTTON-MOMENTARY CONTACT NORMALLY OPEN
PUSH BUTTON LOCK-OUT
PUSH BUTTON-MAINTAINED CONTACT
49
CONTACTS OF OVERLOAD DEVICES
OFF
SELECTOR SWITCH (TWO OR THREE POSITIONS)
SOLENOID
NORMALLY OPEN CONTACT (NO)
NORMALLY CLOSED CONTACT (NC)
MOTOR OPERATED VALVE POSITION LIMIT SWITCH
MOTOR OPERATED VALVE TORQUE SWITCH
B. OPERATING COIL & COMPONENTS
CC — CONTACTOR COIL
CR — CONTROL RELAY
LR — LATCHING RELAY
M — MOTOR STARTER OPERATING COIL
MC — VALVE MOTOR STARTER CLOSE COIL
MO — VALVE MOTOR STARTER OPEN COIL
TDDO — TIME DELAY DROPOUT
TDPU — TIME DELAY PICKUP
42-C — REVERSING MAGNETIC CONTACTOR CLOSING COIL
42-Mx — MOTOR STARTER AUXILIARY RELAY
42-O — REVERSING MAGNETIC CONTACTOR OPENING COIL
52-CC — CIRCUIT BREAKER CLOSE COIL
52-TC — CIRCUIT BREAKER TRIP COIL
OR
C. MISCELLANEOUS DEVICES
CS — CONTROL SWITCH
dPS — DIFFERENTIAL PRESSURE SW
dPSH — DIFFERENTIAL PRESSURE SW
E/P — ELECTRO-PNEUMATIC CONVEI
EP — ELECTRO-PNEUMATIC SWITCH
FM — FLOW METER
FR — FIRE RELAY
FS — FLOW SWITCH
FT — FLOW TRANSMITTER
H — HUMIDISTAT
LMS — LIMIT SWITCH
LS — LEVEL SWITCH
LT — LEVEL TRANSMITTER
PE — PNEUMATIC-ELECTRIC RELAY
PIL — POTENTIAL INDICATING LIGH
PMS — PERMISSIVE SWITCH
PS — PRESSURE SWITCH
PT — PRESSURE TRANSMITTER
RTD — RESISTANCE TEMPERATURE
SD — SMOKE DETECTOR
SS — SELECTOR SWITCH
SV — SOLENOID VALVE CONTACT
T — THERMOSTAT
T/C — THERMOCOUPLE
TE — TEMPERATURE ELEMENT
THD — THERMODETECTOR IN MOTOR
TS — TEMPERATURE SWITCH
OR
INDICATING LIGHT R=RED G=GREEN W=WHITE A=AMBER B=BLUE C=CLEAR
INDICATING LIGHT WITH F (LIGHT COLOR CODE AS ABO
BA
BELL ALARM CONTACT
ANNUNCIATOR POINT
COMPUTER INPUT OR OUTPU
THERMOCOUPLE
RTD
RESISTANCE TEMPERATURE
--E--
ELECTRICAL INTERLOCK
--M--
MECHANICAL INTERLOCK
--K--
KEY INTERLOCK
SINGLE CONDUCTOR WITH SHIELD-UNGROUNDED
SINGLE CONDUCTOR WITH SHIELD-GROUNDED
ABBREVIATIONS
(USED WITH POWER TRANSFORMER
AA — OPEN DRY TYPE
GA — GAS FILLED DRY TYPE
OA — OIL FILLED, SELF COOLED T
FA — OIL FILLED, FORCED AIR CO
FOA — OIL FILLED, FORCED OIL, FO AIR COOLED TYPE

# 1. Introduction and General Description of the Plant

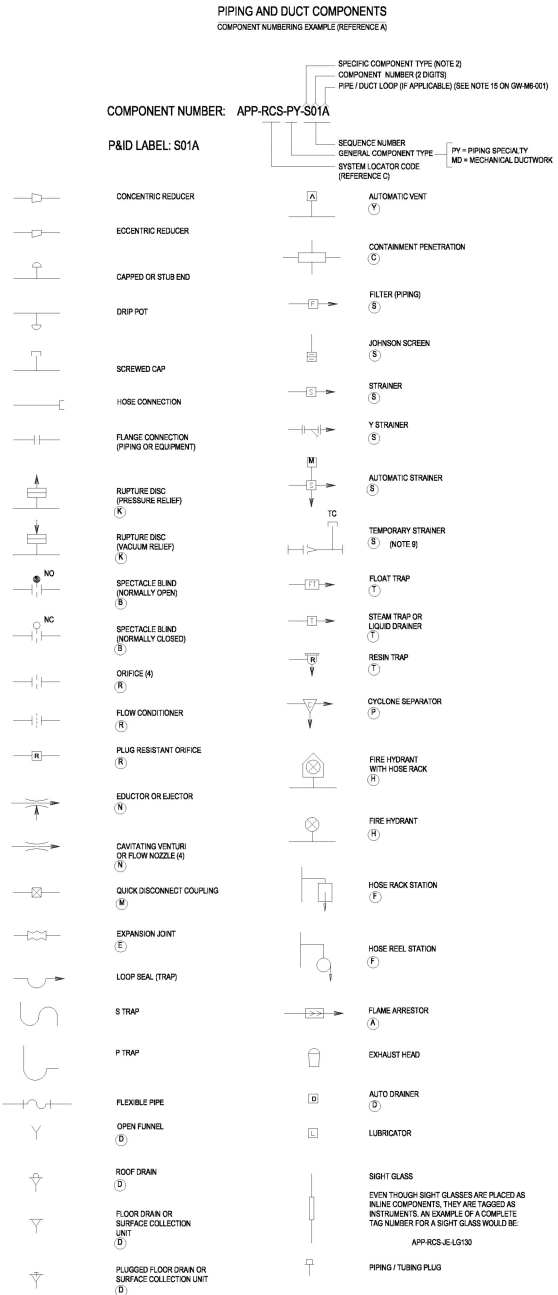


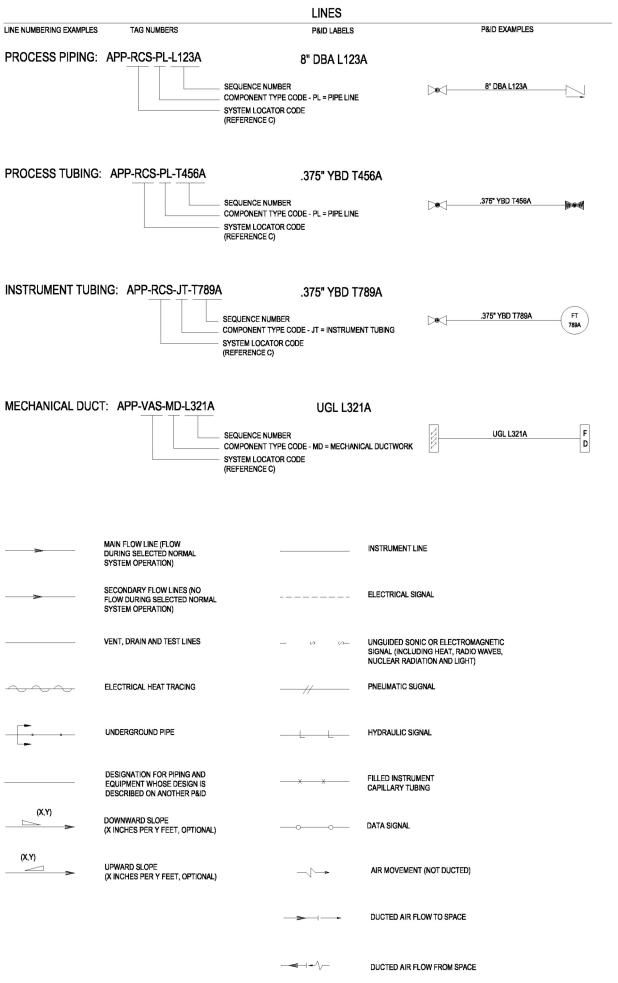
# 1. Introduction and General Description of the Plant



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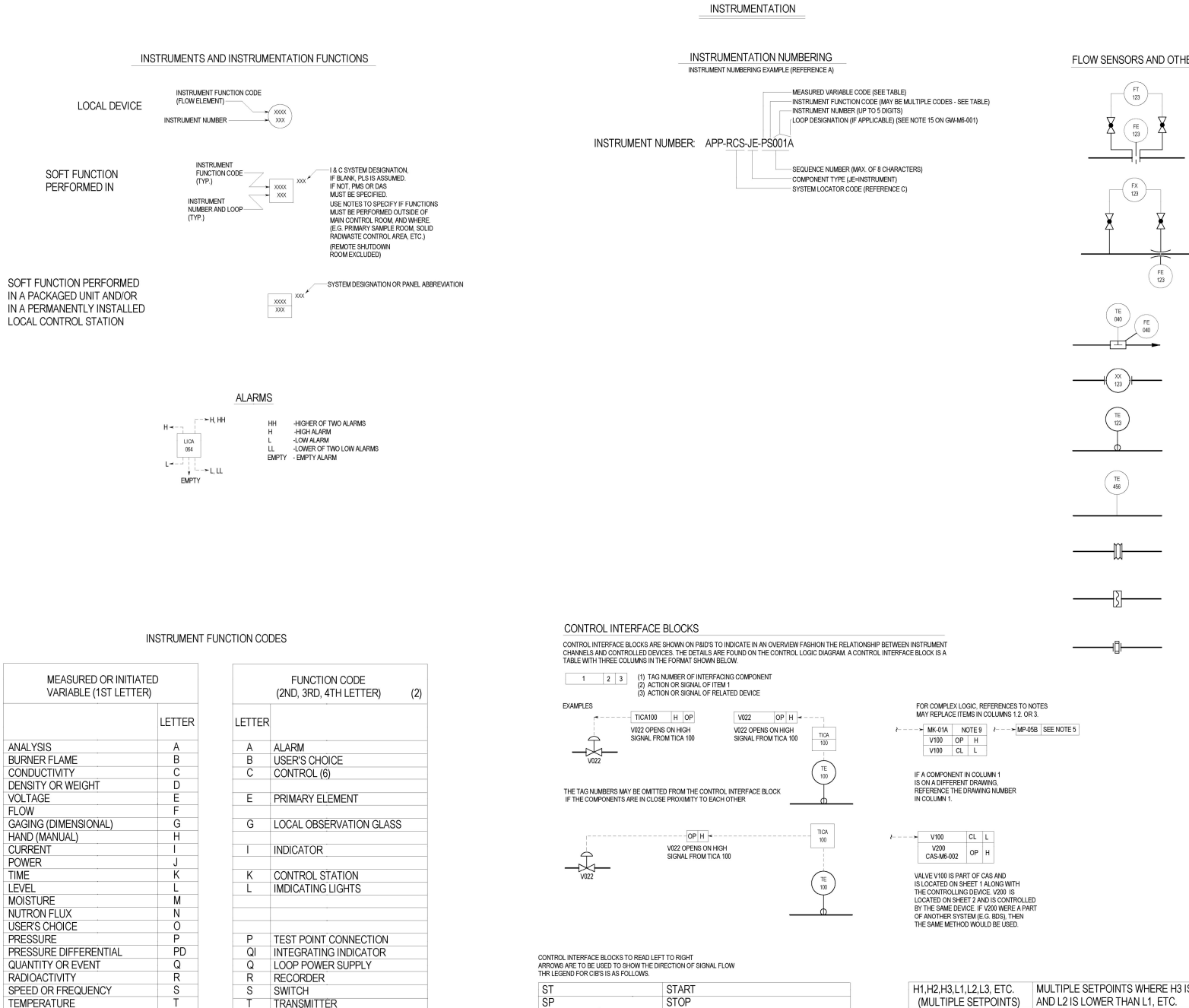



LINE AND VALVE / DAMPER SPECIFICATIONS

| CODE | NOMINAL PRESSURE RATING (FIRST LETTER) | BASIC PIPE LINE MATERIAL (SECOND LETTER) | DESIGN CODE / STANDARD (THIRD LETTER) |
|---|---|---|---|
| A | CL 2500 ASME B16.5 | ALLOY STEEL, STANDARD | ASME SEC.III, CLASS 1, NRC QUAL. GRP. A |
| B | SPECIAL RATING - SEE CLASS SHEET | AUSTENITIC STAINLESS STEEL, STANDARD | ASME SEC.III, CLASS 2, NRC QUAL. GRP. B |
| C | CL 1500 ASME B16.5 | CARBON STEEL, STANDARD | ASME SEC.III, CLASS 3, NRC QUAL. CRP. C |
| D | CL 900/600 ASME B16.5 | DUCTILE IRON - CEMENT LINED | ASME B31.1, (PIPE)/SMACNA (DUCT) NRC QUAL. GRP. D |
| E | CL 600/900 ASME B16.5 | DUCTILE IRON | ASME B31.1, NON Q |
| F | CL 400 ASME B16.5 | FERROUS, CAST IRON | NATIONAL FIRE PROTECTION ASSOCIATION CODE, FIRE PROTECTION QA |
| G | CL 300 ASME B16.5 | CARBON STEEL, GALVANIZED | NATIONAL FIRE PROTECTION ASSOCIATION CODE |
| H | CL 250 ASME B16.1 | COPPER - NICKEL ALLOY | |
| J | CL 150 ASME B16.5 | PE3408/PE3608 HDPE | |
| K | CL 125 ASME B16.1 | COPPER, BRASS OR BRONZE | |
| L | 150 PSI, CWP | LINED PIPE | SMACNA DUCT CONSTRUCTION STANDARDS |
| M | 125 PSI, CWP | ALUMINUM | |
| N | 100 PSI, CWP | ALLOY 20 (UNS N08020) | |
| P | 80 PSI, CWP | PLASTIC AND NON-METALIC | NATIONAL PLUMBING CODE |
| Q | NOT USED | CARBON STEEL, RUBBER LINED | |
| R | 250 PSI, CWP | CONCRETE | ASME/ANSI AG-1 (ASME NS09) DUCT STANDARD |
| S | 200 PSI, CWP | ALLOY STEEL, HIGH STRENGTH | |
| T | 300 PSI SERVICE (VICTAULIC PRESSFIT) | AUSTENITIC STAINLESS STEEL, HIGH STRENGTH | |
| U | 0" - 2" W.G. | CARBON STEEL, IMPACT TESTED | |
| V | 2" - 6" W.G. | PE4710 HDPE | |
| W | 6" - 10" W.G. | | AMERICAN WATER WORKS ASSOCIATION STANDARD |
| X | OVER 10" W.G. | | |
| Y | SPECIAL USE AS SHOWN ON PIPING CLASS SHEETS | SPECIAL USE AS SHOWN ON PIPING CLASS SHEETS | SPECIAL USE AS SHOWN ON PIPING CLASS SHEETS |

# 1. Introduction and General Description of the Plant




| MEASURED OR INITIATED VARIABLE (1ST LETTER) | LETTER |
|---|---|
| ANALYSIS | A |
| BURNER FLAME | B |
| CONDUCTIVITY | C |
| DENSITY OR WEIGHT | D |
| VOLTAGE | E |
| FLOW | F |
| GAGING (DIMENSIONAL) | G |
| HAND (MANUAL) | H |
| CURRENT | I |
| POWER | J |
| TIME | K |
| LEVEL | L |
| MOISTURE | M |
| NUTRON FLUX | N |
| USER'S CHOICE | O |
| PRESSURE | P |
| PRESSURE DIFFERENTIAL | PD |
| QUANTITY OR EVENT | Q |
| RADIOACTIVITY | R |
| SPEED OR FREQUENCY | S |
| TEMPERATURE | T |

| FUNCTION CODE (2ND, 3RD, 4TH LETTER) (2) | |
|---|---|
| LETTER | |
| A | ALARM |
| B | USER'S CHOICE |
| C | CONTROL (6) |
| | |
| E | PRIMARY ELEMENT |
| | |
| G | LOCAL OBSERVATION GLASS |
| | |
| I | INDICATOR |
| | |
| K | CONTROL STATION |
| L | IMDICATING LIGHTS |
| | |
| | |
| | |
| P | TEST POINT CONNECTION |
| QI | INTEGRATING INDICATOR |
| Q | LOOP POWER SUPPLY |
| R | RECORDER |
| S | SWITCH |
| T | TRANSMITTER |